



FDK™ Toolkit, Justice Domain Fairness Audit

External Validation Report: COMPAS Recidivism Dataset

First Real-World Benchmark Validation Following Synthetic Diagnostic Remediation

Dataset: ProPublica COMPAS Recidivism Risk Score dataset (compas_dataset.csv), 7,214 defendants, Broward County, Florida, 2013–2014

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Abstract

This report documents the first validation of the FDK™ Toolkit's Justice fairness-audit pipeline against a real, publicly available benchmark dataset, the ProPublica COMPAS recidivism risk-score dataset, following a comprehensive prior remediation programme conducted entirely on synthetic diagnostic data. The initial validation run surfaced two genuine defects that had not manifested during synthetic testing: an all-or-nothing subgroup-size validation rule that rejected the entire 7,214-row dataset over a single small demographic group, and a missing-value handling gap in the feature-attribution-bias calculation that produced an undefined (NaN) result when a passthrough feature column was absent for the dataset as a whole. Both defects were root-caused, fixed, and verified through iterative re-testing. The Justice pipeline now produces a complete, numerically valid, 35-metric fairness audit of the COMPAS dataset, detecting a statistically significant and substantively large racial disparity in predicted recidivism risk consistent with COMPAS's well-documented public record. This constitutes the toolkit's first successful external, real-world validation and a template for the remaining six domains' equivalent validation work.

1. Background and Purpose

Following a comprehensive internal remediation programme, during which sixteen distinct defects were identified and resolved across all seven FDK™ Toolkit domain pipelines using purpose-built synthetic diagnostic datasets with hand-computed oracle values, the toolkit's Standard Operating Procedure called for a subsequent phase of validation against real or well-established public benchmark datasets to confirm that fixes verified on synthetic data generalise to genuine, messy, real-world input.

The Justice domain was selected for this first real-dataset validation using the ProPublica COMPAS dataset: a widely cited, extensively studied criminal-justice risk-assessment dataset that has itself been the subject of substantial public fairness scrutiny since ProPublica's original 2016 investigation. Its use here serves two purposes: confirming the FDK™ Toolkit's own correctness against real data, and reproducing a well-known fairness finding as an external sanity check on the toolkit's output.

2. Dataset Description

Property	Value
Total records	7,214 defendants
Screening date range	1 January 2013 – 31 December 2014
Group column used	race (6 categories)
Ground-truth column	two_year_recid (actual two-year reoffense, binary)
Prediction column	is_recid (COMPAS system's own recidivism flag, binary)

Property	Value
Probability/risk-score column	Not mapped in this run (decile_score is an ordinal 1–10 field, not a continuous 0–1 probability; see Section 5)
Race group sizes	African-American: 3,696 · Caucasian: 2,454 · Hispanic: 637 · Other: 377 · Asian: 32 · Native American: 18

Two properties of this real dataset are directly relevant to the findings in Section 3: one demographic group (Native American, n=18) falls below the toolkit's minimum-subgroup-size threshold of 20 samples used elsewhere in this project; and one passthrough numeric column present in the raw file, violent_recid, is missing for all 7,214 records without exception (i.e., entirely unpopulated, not merely sparse).

3. Findings From Real-Dataset Validation

Finding 1, All-or-nothing subgroup rejection

The initial validation run rejected the dataset in its entirety:

```
"metrics_calculated": 0,
"error": "Smallest subgroup(s) below 20 samples (< 20 required for
statistical validity): {'Native American': 18}"
```

This guard was originally introduced during the toolkit's remediation programme to prevent degenerate results from genuinely unusable subgroups (the precedent case being a subgroup of a single sample). Applied without qualification, however, it meant that a single small-but-real demographic group, 18 records out of 7,214, or 0.25% of the dataset, was sufficient to block analysis of the other 99.75% entirely, including two groups (African-American, n=3,696; Caucasian, n=2,454) with very strong statistical power. This is a materially different situation from the original 1-sample case that motivated the rule, and treating them identically was assessed as overly conservative for a real-world auditing context, where small-but-genuine demographic groups are common and should not disqualify an otherwise well-powered audit.

Remediation: the hard rejection was replaced with a soft-exclusion policy. Groups below the 20-sample threshold are now excluded from the analysis, with the exclusion and its reason recorded transparently in the output; the audit proceeds on the remaining adequately-sized groups. A hard rejection is retained only for the genuinely degenerate case of fewer than two groups remaining after exclusion.

Finding 2, Undefined result from a fully-missing feature column

Following the fix above, the pipeline completed but returned an undefined numeric value for one metric:

```
"feature_attribution_bias": NaN
```

Root cause: the feature-attribution-bias calculation incorporates every numeric column not already mapped to a standard role (group, ground truth, prediction, probability) as a candidate feature for cross-group comparison. In the COMPAS dataset, the violent_recid column is present in the schema but contains no non-missing values for any of the 7,214 records. The calculation's per-group mean for this column was therefore undefined for every group, and this single undefined value propagated through the metric's final averaging step, invalidating the entire result regardless of how well populated the other candidate features were.

Remediation: the calculation now detects, per feature column, whether any individual group has zero non-missing values for that feature, and excludes only that column from the comparison in that case, rather than allowing one entirely-missing column to invalidate the metric as a whole. All other genuinely populated features (age, prior offence counts, COMPAS decile score, etc.) continue to contribute normally.

Verification of both fixes

Metric	Before fix	After fix	Result
metrics_calculated	0 (rejected entirely)	35 / 35	✓ Resolved
feature_attribution_bias	NaN	0.4308	✓ Resolved
data_quality_warnings	not present	present, names excluded group and reason	✓ Transparent
Confusion-matrix metrics (fpr/tpr/etc.)	n/a (audit did not run)	real, group-varying values	✓ Verified

4. Fairness Findings on the COMPAS Dataset

With both defects resolved, the Justice pipeline produced a complete 35-metric audit across five race groups (African-American, Caucasian, Hispanic, Other, and Asian; Native American excluded per Finding 1 above). Headline results:

Metric	Value	Interpretation
disparate_impact.ratio	0.624	Below the conventional 0.8 four-fifths threshold; flagged HIGH severity by the toolkit
statistical_parity_difference	0.207	A 20.7 percentage-point gap in predicted-positive (flagged as high-risk) rate between the highest and lowest group
selection_rates_by_group (African-American)	0.551	55.1% flagged as high recidivism risk by the underlying algorithm
selection_rates_by_group (Caucasian)	0.418	41.8% flagged as high recidivism risk
selection_rates_by_group (Asian)	0.344	Lowest predicted-positive rate of the five analysed groups
fpr_ratio	0.369	The most disadvantaged group's false-positive rate is roughly 2.7× the least-disadvantaged group's
overall_assessment	HIGH_BIAS	Toolkit's own composite classification of the result

These findings are directionally consistent with ProPublica's original, widely cited 2016 analysis of this same dataset, which found that COMPAS's risk scores were more likely to falsely flag Black defendants as high-risk than white defendants, at a materially higher rate. The FDK™ Toolkit's independent, automated audit, computed via genuinely distinct pipeline logic, reproduces that same qualitative direction of disparity, which serves as an external, real-world sanity check on the toolkit's correctness beyond the synthetic oracle-verification already completed.

5. Scope Limitations of This Validation Run

- No probability/risk-score column was mapped for this run. COMPAS's native `decile_score` field is an ordinal 1–10 ranking, not a continuous 0–1 probability, and the toolkit's current `y_prob` detection requires the latter. As a result, calibration and AUC-based metrics (`slice_auc_difference` and related fields) did not populate meaningfully for this run. A rescaled `decile_score` (e.g. $(\text{decile_score} - 1) / 9$) would allow these metrics to run in a future validation pass, understanding that this is a linear-rescaling approximation of an inherently ordinal score, not a native probability.
- `temporal_fairness_score` (0.111) is notably lower than typical values seen during synthetic diagnostic testing. This reflects COMPAS's genuine, uneven real-world screening-date distribution across a two-year window, rather than a defect; it has not been independently investigated further as part of this validation and is noted here for completeness.
- This validation covers the Justice domain only. The remaining six FDK™ Toolkit domains have not yet undergone equivalent real-dataset validation; the subgroup soft-exclusion fix verified here is a candidate for extension to all seven domains as a consistency measure.

6. Conclusion

The Justice domain fairness-audit pipeline has been successfully validated against a real, independently collected, publicly scrutinised benchmark dataset. Two genuine defects, both real-world data-handling gaps that synthetic testing could not have surfaced, since they depend specifically on demographic group-size imbalance and column-level missingness patterns intrinsic to real data, were identified, root-caused, fixed, and verified. The pipeline's resulting fairness findings on COMPAS are directionally consistent with established public research on this dataset, providing external corroboration of the toolkit's correctness beyond its own internal oracle-based verification. The Justice domain is assessed as validated for real-world use, with the scope limitations in Section 5 noted for future work.

Appendix, Reference Materials

Dataset: `compas_dataset.csv` (ProPublica COMPAS Recidivism Risk Score dataset, Broward County, FL, 2013–2014)

Prior reports: `FDK_Toolkit_Findings_Remediation_Report.docx`; `FDK_Toolkit_Production_Readiness_Report.docx`

Modified source file: `Justice/fdk_justice_pipeline.py` (`calculate_all_metrics` subgroup handling; `_calculate_feature_attribution_bias` missing-value handling)

Validation reports referenced: `justice_audit_report_20260721_155510.json` (post subgroup-fix, pre-feature-attribution-fix); `justice_audit_report_20260721_160350.json` (fully resolved)